

A Serious Challenge to a New Building Technology

Every new building technology comes with new problems along with its benefits, and structural insulated panels (SIPs) are no different. SIPs are made of rigid foam insulation sandwiched between sheets of oriented strand board (OSB). The sheets of OSB act as load-bearing members, so there is no need for framing, which saves wood. SIPs can produce a very tight and well-insulated building envelope when properly sealed (see “The Lowdown on Structural Insulated Panels,” *HE* Jan/Feb ’02, p. 38). But therein lies the problem. When the joints are not properly sealed, air and water vapor infiltration, especially in cold climates, can lead to disaster.

The catastrophic failure of 109 SIPs-built residential units in Juneau, Alaska highlights the challenges associated with building with SIPs. Failures that I’ve identified since March 2001 include roof and wall failures; the structures involved range from single-family houses to 15-unit apartment buildings. A study of 37 roof failures pointed to poor installation as the major culprit (see “A Problem with Poor Installation?”). But as a project manager in identifying the 109 failures and in erecting 21 replacement roofs, I’ve discovered problems with SIP manufacturing as well.

Hard Lessons

One of the failures of Premier Industry panels in Juneau was on a project where the manufacturer had a representative on-site at least part-time. All but three of the units that were built with Insulspan panels that failed were installed by factory crews or under factory supervision. Several contractors were involved in the actual building of the failed units in Juneau; some of these were trained by the manufacturers in proper procedures.

Panels that had apparently good seals failed in the same fashion as panels that had obvious construction defects. The installation defects compounded the design and manufacturing defects in the panel system joints. The manufacturing problems with SIPs include out-of-square panels and routings for the



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(top) Fungus growing on a SIPs roof is a sure sign of trouble. (middle) The author is exposing a failed joint. (bottom) This failed joint had an apparently good sealant installation.

splines that are significantly larger than the spline material, making it impossible to achieve an effective seal. The seal is contained inside the panel skins, which are connected with splines that completely hide the sealant. Once the panels are in place, visual inspection would reveal only gross defects in the seal. When the sealant is compressed against the foam core rather than being bonded to the OSB on both sides of the joint, water vapor can pass through microscopic voids. Thus, failure is not detected until it reaches catastrophic proportions, as happened in Juneau.

The manufacturer's sealing instructions provide for a one-shot bar-

rier to moisture penetration. To be effective, the barrier must keep water vapor on the warm side of the dew point. Moisture vapor can penetrate the joints as long as the outside temperature is at or below the dew point of the inside air. Locations that have less cool weather than in Juneau could be expected to experience the same type of damage—but not as quickly. Juneau has about 95% cool or cold weather, but my company recently received photographs of a house in central Ohio that had a damaged SIP roof and walls. The house is the same age as the average unit in Juneau, and the panels appeared to be about half as damaged as typical units in Juneau. According to the information we received, the panels were manufactured by one of the manufacturers involved in Juneau and were erected by a related company.

Other factors, such as occupant load and solar exposure, also affect the rate of damage by affecting the amount of moisture in the inside air and the rate of condensation in the SIPs. The manufacturers have suggested that the absence or inadequate functioning of the heat recovery ventilators (HRV) is to blame for the damage observed, but an HRV is designed primarily to ensure occupant comfort; it is not meant to protect the structure.

What Will Help?

Building Science Corporation's Joe Lstiburek noted in a report on Juneau roof failures that the makers of Insulspan panels, which were on the roofs that his people inspected, needed to redesign their joints. The report stated, “New joint designs can be developed that rely on gaskets and interlocking joints.” We suggest that they also need to redesign their system to include

- a continuous vapor retarder of significantly lower perm rating than the OSB skin on the warm surface of the panels, which in Juneau is the inside surface;

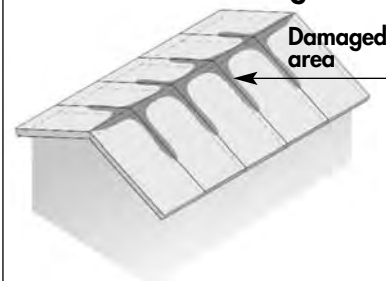
- specific instructions on how to achieve effective seals in the panel joints with the provided sealants, including weather, temperature, and surface mois-

A Problem with Poor Installation?

Bill Wachler, the executive director of the Structural Insulated Panel Association (SIPA), hired Building Science Corporation's Joe Lstiburek to investigate roof failures in 37 houses in Juneau. Lstiburek found that the roof failures were caused by the failure of SIPs joints (see Figures A and B). Warm, moist indoor air migrated through improperly sealed joints inside the houses and condensed on the colder upper OSB edges on the roof peak, wetting the joint surfaces and building paper. The water spread through capillary action between the upper surface of the OSB and the roofing paper. This led to rotting of the roof along the joints.

Lstiburek found that in some places, no effort was made to seal the panel joints. (SIPA recommends the use of high-yield expanding foam and/or an adhesive specially formulated for SIPs construction.) In other places, sealant was applied in a haphazard and ineffective manner, or the sealant failed to adhere properly. The first two cases are clearly a case of poor workmanship. Lstiburek recommended fixes that mostly

The Pattern of Damage



Moisture Entry and Redistribution Within Panel Joints

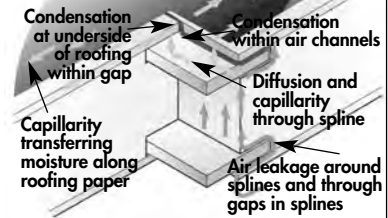


Figure A. (left) The decay of the SIPs roofs followed the panel joints. The roof ridges and peaks were the most effected. **Figure B.** (right) In the cold and wet Juneau climate, water vapor allowed into the SIPs panel joints from the warm side of the roof will condense on the cold underside of the roofing paper.

centered on stopping the air flow from the inside of the SIPs roofs. Regarding the third category—failure due to poor sealant cohesion—the report states, “The difficulty of applying sealants in the appropriate location under typical weather conditions in Juneau, Alaska is, in our opinion, a major issue.”

The SIPA chapter in Alaska, in an attempt to correct the problems that the roof failures revealed, is working with homebuilders to develop a certification process to ensure proper panel installation. According to Wachler, every SIP

manufacturer has clear drawings and installation instructions on how to build for a particular climate. Wachler sees the problem as one of getting the installers to follow these instructions. In the Juneau cases, the primary panel manufacturers were Insulspan and Premier Industry. Wachler claims that the contractor they worked with subcontracted the work to installers who had no experience with panels.

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ture limitations, and exact placement instructions;

- detailed instructions on the installation of the vapor retarder and design of the heat recovery ventilator (HRV) system; and
- a cold-roof design that is mandatory.

A properly installed vapor retarder of significantly lower permeability than the OSB serves as a first line of resistance to vapor or air penetration of the panel joints. Properly installed sealants, adhered to both adjacent surfaces at the proper locations, provide a second line of resistance to moisture penetration to the level in the joint where condensation can occur. A cold roof installed over the panels provides a damage control mechanism in the event that the first two mechanisms fail.

While we would never sanction depending on the HRV to protect the structure, a properly designed HRV can be beneficial to the structure by preventing the accumulation of warm moist air

in the upper levels of the structure, and by maintaining a slight negative pressure on the structure, which helps to minimize the vapor pressure. Designing an HRV to benefit the structure requires increasing air flow and changing inlet/outlet locations—a more costly alternative than one that simply addresses occupant comfort. Over the life of a structure, it is unlikely that the vapor retarder will remain intact. Multiple layers of vapor resistance, comparable to the multiple load paths in structural engineering, provide the best chance for survival of the SIP structure.

In many respects, Juneau is the Challenger disaster of the SIP industry and, to a lesser extent, of code certification agencies such as Building Officials and Code Administrators International, Incorporated (BOCA), and the International Conference of Building Officials Evaluation Service (ICBO ES). The SIPs were tested and certified for their structural performance, but apparently they were never evaluated by the code

agencies in a system-oriented process that would have tested the joint design. However, the code agency certifications were interpreted to cover all of the manufacturer's installation instructions and recommendations, not just the structural issues, and the process leading to the failures was irreversibly set in motion. Whether the SIP industry learns its lessons and applies them as NASA did is the big question confronting the industry and the buying public.



—John D. Cooper

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